

MCSL-229 (Set-4)
MASTER OF COMPUTER APPLICATIONS
(MCA-NEW)
Cloud and Data Science Lab

Duration : 2 hours

Maximum Marks : 50

- Note:
1. There are two sections in this paper. Each section is of one hour duration.
 2. Each section has one compulsory question of 20 marks. Each section has 5 marks for viva voce.
 3. Attempt only those section(s) in which you are not successful as yet.

SECTION-A

(Cloud Computing Lab)

1. (a) Create a document of your examination schedule using Google Docs and store locally on a Google drive with doc and pdf formats. Share this document in view mode. 10
- (b) List the features of Just Cloud file storage and sharing solutions. 10

SECTION B
(Data Science Lab)

2. Consider the following data about study time in a day and final percentage of marks of 10 students: 20

Student Number	Study time in Hours (in a day)	Final Percentage of Marks
1	5	75%
2	6	75%
3	3	60%
4	4	62%
5	1	55%
6	2	58%
7	8	80%
8	6	75%
9	7	80%
10	4	60%

Use R programming to fit a linear regression line to predict the effect of study time on the final percentage of the students. Use this regression line to predict the final percentage of a student who studies for 5 hours every day.

MCSL-229 (Set-3)
MASTER OF COMPUTER APPLICATIONS
(MCA-NEW)
Cloud and Data Science Lab

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SECTION-A
(Cloud Computing Lab)

1. (a) Use Google Slides and prepare 5 slides on the topic “Cloud Computing”.
Share these slides in a group in edit mode. 10
- (b) List the features of Oracle Virtual box. Also, list the steps of creating
Virtual Machine (VM) on your computer. 10

SECTION B

(Data Science Lab)

2. Write R program for the following income data of 10 persons: 20

25,000	35,000	30,000	45,000	29,000
27,000	47,000	51,000	25,000	39,000

Write a R program to create a frequency distribution in the three classes:

20,000 – 30,000

30,000 – 40,000

40,000 – 50,000

Also write R program to draw appropriate chart for the frequency distribution.

MCSL-229 (Set-2)
MASTER OF COMPUTER APPLICATIONS
(MCA-NEW)
Cloud and Data Science Lab

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SECTION-A
(Cloud Computing Lab)

1. (a) Use Google docs and create a document consisting of information about cloud computing. The document should include proper headings and page number. Store this document on the google drive and provide permission to view this document. 10
- (b) Create a list of cloud services provided by AWS. List the steps to setup an Elastic compute cloud (EC2) instance. 10

SECTION-B

(Data Science Lab)

2. The following data was collected to predict the weight of a person from his/her height: 20

Height (cm)	160	155	175	163	171	183	159	162
Weight (Kg)	65	60	75	65	73	85	60	65

Use R programming to fit a linear regression line to predict the weight of a person using his/her height. Also, predict the weight of a person whose height is 165 cms.

MCSL-229 (Set-1)
MASTER OF COMPUTER APPLICATIONS
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Cloud and Data Science Lab

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SECTION-A
(Cloud Computing Lab)

1. (a) Create a spreadsheet using Google sheets that contains employee salary information and calculate net salary. You may use the following labels and data:

10

Name	Basic Salary	DA	PF	TAX	Net Salary
ABC	50,000				
XYZ	25,000				
DEF	75,000				

You need to compute DA, PF, TAX and Net Salary using the following formula:

DA = 20% of the Basic Salary

PF = 10% of Basic Salary if Basic salary $\leq$ 30,000

12% of Basic Salary if Basic salary $>$ 30,000

TAX = 10% of Basic Salary if Basic salary $<$ 50,000

TAX = 15% of Basic Salary if Basic salary $\geq$ 50,000

Net Salary = Basic salary + DA + PF – TAX

- (b) List the feature of YouTube, a cloud Service. List the steps of uploading your own educational video(s) on YouTube. Also, list the appropriate settings to make it public. 10

SECTION B

(Data Science Lab)

2. The weight of 10 students in the age group 15-20 is given in the following table:

45	55	65	38	48
50	54	60	39	49

Write R program (use the data given below) for the following:

- (i) Finding the minimum and maximum weight. 4
- (ii) Create a grouped frequency distribution and relevant graph of frequency distribution for the data given in the table above. 8
- (iii) Find the percentage of weights between 40 and 49. 8

No. of Printed Pages : 2 **MCSL-229(Set-I)**

**MASTER OF COMPUTER
APPLICATIONS (MCA-NEW)
Term-End Practical Examination
December, 2024
MCSL-229(Set-I) : CLOUD COMPUTING AND
DATA SCIENCE LAB**

Time : 2 Hours

Maximum Marks : 50

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- Note :** (i) *There are two Sections in this paper.*
- (ii) *Each Section is of one hour duration.*
- (iii) *Each Section has one compulsory question of 20 marks.*
- (iv) *Each Section has 5 marks for viva-voce.*
- (v) *Attempt only those Section(s) in which you are not successful as yet.*
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**Section—A
(Cloud Computing Lab)**

1. (a) Use Google DOCs and create a resume with appropriate formatting. Include sections like education, skills and contact

[2]

information. Store this document on the Google drive and provide permission to view this document. 10

- (b) Discuss the process of creating and deploying a Google App Engine application. 10

Section—B
(Data Science Lab)

2. Consider the following data of students' daily revision time (in hours) and their final exam scores. Use linear regression in R to predict the final score for a student who revises for 6 hours a day : 20

Revision Time (hours)	Final Score (%)
2	50
4	60
6	75
8	85
10	90

× × × × × × ×

No. of Printed Pages : 2 **MCSL-229(Set-IV)**

**MASTER OF COMPUTER
APPLICATIONS (MCA-NEW)
Term-End Practical Examination
December, 2024**

**MCSL-229(Set-IV) : CLOUD COMPUTING AND
DATA SCIENCE LAB**

Time : 2 Hours

Maximum Marks : 50

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- Note :** (i) *There are two Sections in this paper.*
(ii) *Each Section is of one-hour duration.*
(iii) *Each Section has one compulsory question of 20 marks.*
(iv) *Each Section has 5 marks for viva-voce.*
(v) *Attempt only those Section(s) in which you are not successful as yet.*
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Section—A

(Cloud Computing Lab)

1. (a) Create a Google sheet document to store employee information with the following columns :
Name, Basic Salary, DA, HRA, PF, Tax, Gross Salary, Net Salary
Use the following formulas to calculate : 10
DA = 12% of Basic
HRA = 25% of Basic

[2]

PF = 10% of Basic (if Basic $\leq$ 5000; 12%
(if Basic > 5000)

Tax = 80% of Basic

Gross Salary = Basic + DA + HRA

Net Salary = Gross Salary – PF – Tax

- (b) List and compare the storage and sharing features of the following cloud services :

10

Box, Sync, Just Cloud and Amazon Drive.

Section—B

(Data Science Lab)

2. The following data shows study hours and grades of students. Use linear regression to predict the grade for a student who studies for 9 hours a week :

20

Study Hours	Grade (%)
2	40
4	55
6	65
8	80
10	90

× × × × × × ×

No. of Printed Pages : 2 **MCSL-229(P)Set-IV**

**MASTER OF COMPUTER
APPLICATIONS (MCA-NEW)
Term-End Practical Examination
June, 2025**

**MCSL-229(P)Set-IV : CLOUD AND DATA
SCIENCE LAB**

Time : 2 Hours

Maximum Marks : 50

Note : (i) *There are two Sections in this paper.
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question of 20 marks. Each Section
has 5 marks for viva-voce.*

(iii) *Attempt only those Section(s) in
which you are not successful as yet.*

Section—A : Cloud Computing Lab

1. (a) Design a Google form for feedback collection after a Webinar. Include questions using different types like

[2]

short answer, multiple choice, checkboxes and linear scale. Share it and collect responses. 10

- (b) Describe the concept of SaaS (Software as a Service). Give examples and mention advantages over traditional software models. 10

Section—B : Data Science Lab

2. Daily sales (in units) for a store in 10 days are recorded as :

150, 160, 145, 170, 155, 165, 158, 172, 169, 160

Write R program to perform the following tasks :

- (a) Plot a histogram of daily sales. 4
(b) Compute and interpret the mode. 8
(c) Test whether the data is normally distributed using appropriate method in R. 8

x x x x x

No. of Printed Pages : 2 **MCSL-229(P)Set-III**

**MASTER OF COMPUTER
APPLICATIONS (MCA-NEW)
Term-End Practical Examination
June, 2025**

**MCSL-229(P)Set-III : CLOUD AND DATA
SCIENCE LAB**

Time : 2 Hours

Maximum Marks : 50

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question of 20 marks. Each Section
has 5 marks for viva-voce.*

(iii) *Attempt only those Section(s) in
which you are not successful as yet.*

Section—A : Cloud Computing Lab

1. (a) Create a presentation using Google slides explaining the architecture of

cloud computing. Include at least *five* slides with text, images and transitions.

10

- (b) What is Amazon Web Services (AWS) ?
List at least *five* major services offered and briefly describe their use. 10

Section—B : Data Science Lab

2. A sample of daily temperature (in °C) recorded over 10 days is as follows :

30, 32, 28, 31, 33, 29, 34, 35, 30, 32

Write R program to perform the following tasks :

- (a) Plot the line graph for daily temperatures. 4
- (b) Calculate the moving average of the last 5 days. 8
- (c) Identify days with temperature above the overall average. 8

× × × × ×

No. of Printed Pages : 2 **MCSL-229(P)Set-II**

**MASTER OF COMPUTER
APPLICATIONS (MCA-NEW)
Term-End Practical Examination
June, 2025**

**MCSL-229(P)Set-II : CLOUD AND DATA
SCIENCE LAB**

Time : 2 Hours

Maximum Marks : 50

Note : (i) *There are two Sections in this paper.
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question of 20 marks. Each Section
has 5 marks for viva-voce.*

(iii) *Attempt only those Section(s) in
which you are not successful as yet.*

Section—A : Cloud Computing Lab

1. (a) Use Google Docs to create a formatted document resembling a research report. Include headings, bullet points,

numbered lists, tables and hyperlinks.

Write a short report on “Benefits of Cloud Computing in Education”. 10

- (b) Explain the benefits of using Dropbox as a cloud storage solution. How can files be shared securely using Dropbox ? Include the steps with examples. 10

Section—B : Data Science Lab

2. Given the data set of exam scores :

72, 85, 91, 68, 77, 81, 95, 89, 73, 88

Write R program to perform the following tasks :

- (a) Calculate the range and interquartile range (IQR). 4
- (b) Create a box plot and interpret the distribution. 8
- (c) Find the proportion of scores greater than 80. 8

× × × × ×

No. of Printed Pages : 3 **MCSL-229(P)Set-I**

**MASTER OF COMPUTER
APPLICATIONS**

(MCA-NEW)

Term-End Practical Examination

June, 2025

**MCSL-229(P)Set-I : CLOUD AND DATA
SCIENCE LAB**

Time : 2 Hours

Maximum Marks : 50

Note : (i) *There are two Sections in this paper.*

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(iii) *Attempt only those Section(s) in which you are not successful as yet.*

Section—A : Cloud Computing Lab

1. (a) Create a spreadsheet using Microsoft Excel that contains student marks

information and calculate total and percentage. You may use the following labels and data : 10

Name	Mark 1	Mark 2	Mark 3	Total	Percentage
Jeevan	78	85	92		
Alia	67	74	69		
Ritu	88	90	85		

You may use below formulas to calculate total and percentage :

$$\text{Total} = \text{Mark 1} + \text{Mark 2} + \text{Mark 3}$$

$$\text{Percentage} = (\text{Total} / 300) \times 100$$

- (b) What is Google drive ? List its features and explain how it facilitates file sharing and collaboration. 10

Section—B : Data Science Lab

2. A data set of monthly electricity bills (in ₹) for 10 households is given :

2200; 1800; 2600; 2100; 2000; 1950; 2300;
1700; 2500; 2400

[3]

Write R program to perform the following tasks :

- (a) Find the mean and median of the electricity bills. 4
- (b) Create a bar chart for the data. 8
- (c) Calculate the standard deviation and interpret it. 8

x x x x x